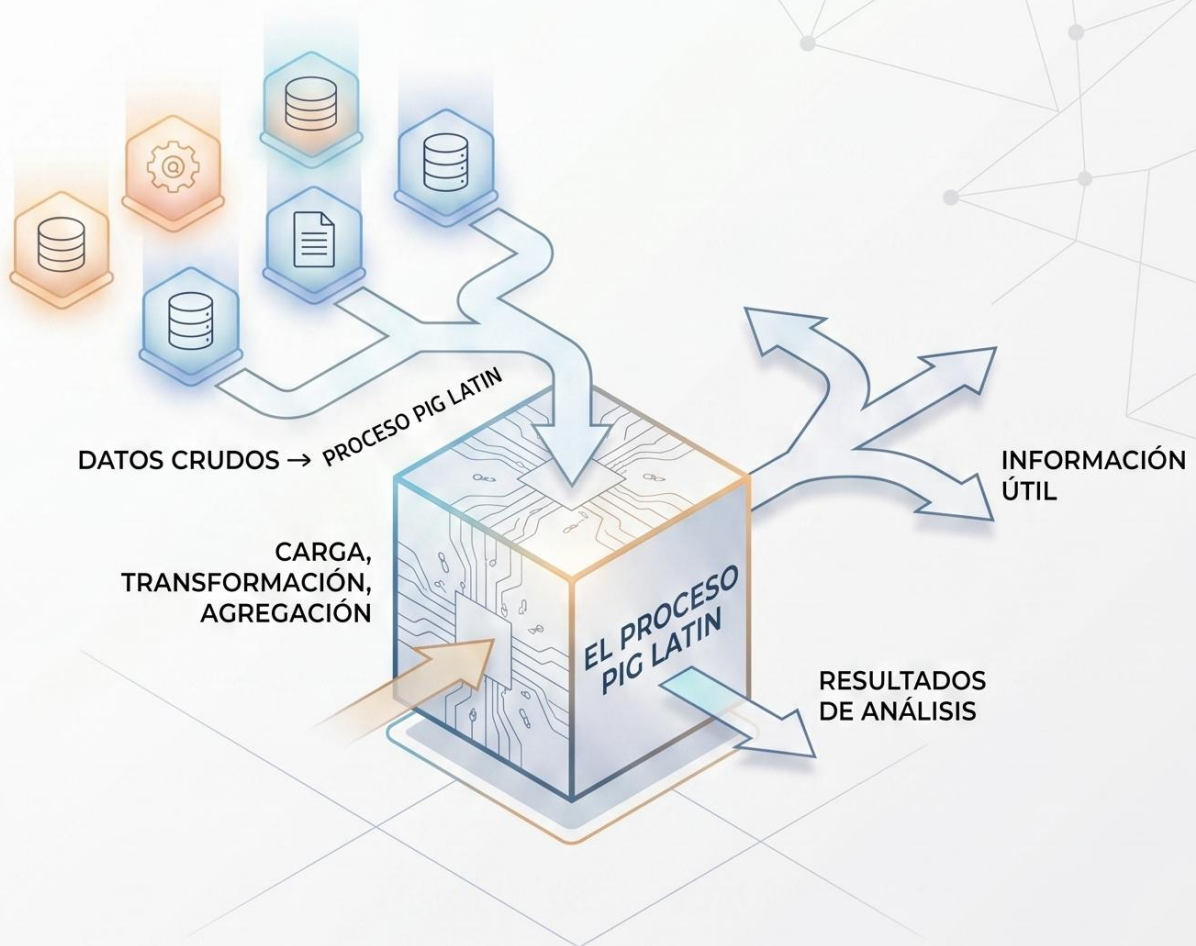




INFORMES DE PRÁCTICAS CON APACHE PIG ENTORNO CLOUDERA VM

IVANA SÁNCHEZ PÉREZ



INGENIERÍA
DE DATOS

Tarea 1: Análisis de Sistemas y Conteo de Estructuras

Introducción

En esta primera fase de la práctica, se establece el contacto inicial con el ecosistema de Hadoop y la herramienta Apache Pig. El objetivo principal es validar la conectividad y el funcionamiento del entorno distribuido en la máquina virtual de Cloudera.

La tarea se centra en la ingesta y procesamiento de datos del sistema local (/etc/passwd), transformándolos en una estructura de datos procesable para realizar un análisis de frecuencias sobre las *shells* de usuario. Este ejercicio sirve como base para comprender el ciclo de "Carga, Agrupación y Almacenamiento" de resultados en el sistema de archivos distribuido HDFS.

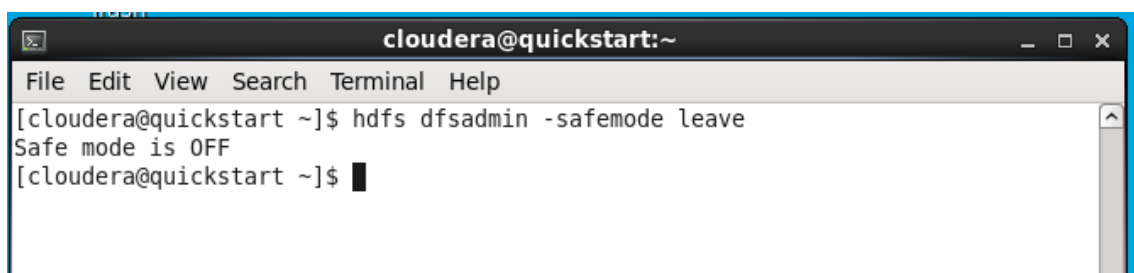
Paso 1: Mover el archivo al "disco duro" de Hadoop (HDFS)

Hadoop no trabaja directamente con los archivos que vemos en nuestras carpetas normales, necesita que se suban a su propio sistema (HDFS).

1. Salir del "Safe Mode"

Hadoop no nos dejará crear carpetas ni subir archivos hasta que desactivemos este modo (el modo seguro) ejecutaremos este comando en el terminal:

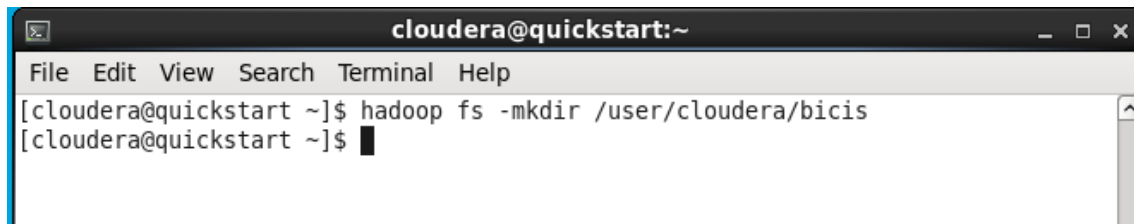
hdfs dfsadmin -safemode leave



```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
[cloudera@quickstart ~]$ hdfs dfsadmin -safemode leave  
Safe mode is OFF  
[cloudera@quickstart ~]$
```

2. En la terminal (sin entrar en pig todavía) crearemos nuestra carpeta de trabajo

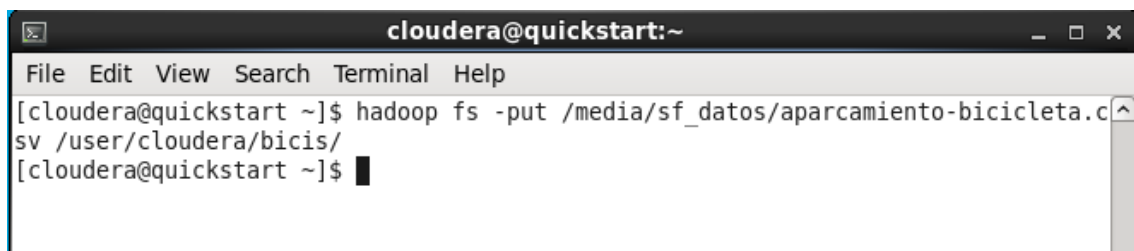
`hadoop fs -mkdir /user/cloudera/bicis`



```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
[cloudera@quickstart ~]$ hadoop fs -mkdir /user/cloudera/bicis  
[cloudera@quickstart ~]$
```

3. Ahora subiremos el archivo CSV a esa carpeta con:

`hadoop fs -put /media/sf_datos/aparcamiento-bicicleta.csv /user/cloudera/bicis/`

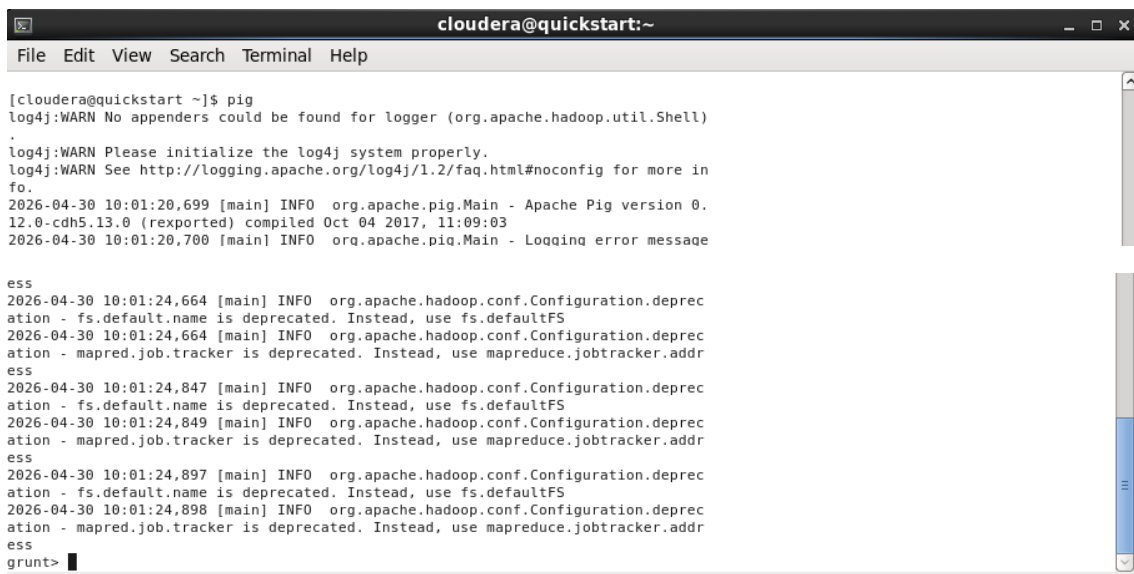


```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
[cloudera@quickstart ~]$ hadoop fs -put /media/sf_datos/aparcamiento-bicicleta.c  
sv /user/cloudera/bicis/  
[cloudera@quickstart ~]$
```

Paso 2: Entrar en Apache Pig

Ahora vamos a entrar en la consola interactiva donde ejecutaremos los comandos de la tarea.

En el terminal escribimos: **pig** para entrar en su entorno (grunt)

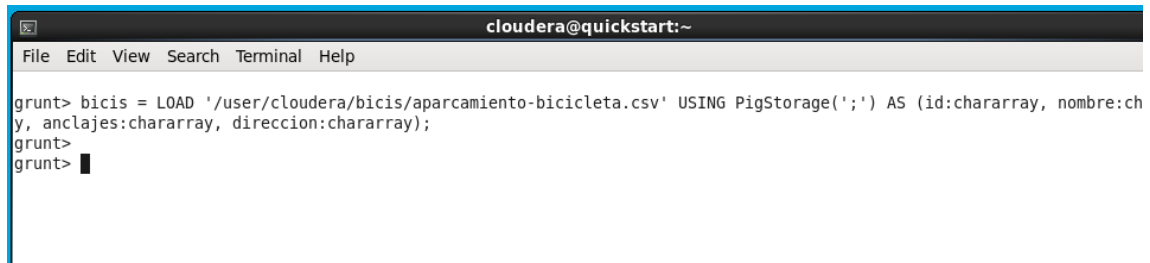


```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
[cloudera@quickstart ~]$ pig  
log4j:WARN No appenders could be found for logger (org.apache.hadoop.util.Shell)  
.  
log4j:WARN Please initialize the log4j system properly.  
log4j:WARN See http://logging.apache.org/log4j/1.2/faq.html#noconfig for more in  
fo.  
2026-04-30 10:01:20,699 [main] INFO org.apache.pig.Main - Apache Pig version 0.  
12.0-cdh5.13.0 (rexpoted) compiled Oct 04 2017, 11:09:03  
2026-04-30 10:01:20,700 [main] INFO org.apache.pig.Main - Logging error message  
ess  
2026-04-30 10:01:24,664 [main] INFO org.apache.hadoop.conf.Configuration.deprec  
ation - fs.default.name is deprecated. Instead, use fs.defaultFS  
2026-04-30 10:01:24,664 [main] INFO org.apache.hadoop.conf.Configuration.deprec  
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr  
ess  
2026-04-30 10:01:24,847 [main] INFO org.apache.hadoop.conf.Configuration.deprec  
ation - fs.default.name is deprecated. Instead, use fs.defaultFS  
2026-04-30 10:01:24,849 [main] INFO org.apache.hadoop.conf.Configuration.deprec  
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr  
ess  
2026-04-30 10:01:24,897 [main] INFO org.apache.hadoop.conf.Configuration.deprec  
ation - fs.default.name is deprecated. Instead, use fs.defaultFS  
2026-04-30 10:01:24,898 [main] INFO org.apache.hadoop.conf.Configuration.deprec  
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr  
ess  
grunt>
```

Paso 3: El Script de la Tarea

1. Cargar los datos

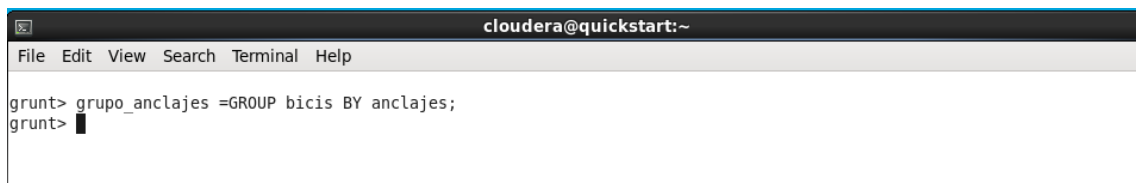
```
bicis = LOAD '/user/cloudera/bicis/aparcamiento-bicicleta.csv' USING
PigStorage(';') AS (id:chararray, nombre:chararray,
anclajes:chararray,
direccion:chararray);
```

A screenshot of a terminal window titled 'cloudera@quickstart:~'. The terminal shows the execution of a Pig script. The prompt is 'grunt>'. The script being entered is: 'bicis = LOAD '/user/cloudera/bicis/aparcamiento-bicicleta.csv' USING PigStorage(';') AS (id:chararray, nombre:ch', followed by a new line, 'y, anclajes:chararray, direccion:chararray);', followed by a new line, and then 'grunt>' and 'grunt>' on separate lines.

```
cloudera@quickstart:~
File Edit View Search Terminal Help
grunt> bicis = LOAD '/user/cloudera/bicis/aparcamiento-bicicleta.csv' USING PigStorage(';') AS (id:chararray, nombre:ch
y, anclajes:chararray, direccion:chararray);
grunt>
grunt>
```

2. Agrupar por número de anclajes: Queremos juntar todas las filas que tengan el mismo número de anclajes.

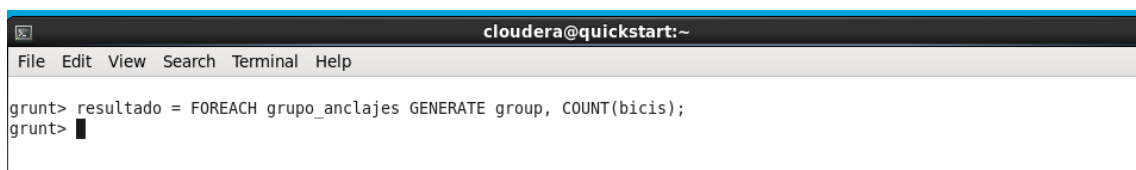
```
grupo_anclajes = GROUP bicis BY anclajes;
```

A screenshot of a terminal window titled 'cloudera@quickstart:~'. The terminal shows the execution of a Pig script. The prompt is 'grunt>'. The script being entered is: 'grupo_anclajes =GROUP bicis BY anclajes;', followed by a new line, and then 'grunt>' on a new line.

```
cloudera@quickstart:~
File Edit View Search Terminal Help
grunt> grupo_anclajes =GROUP bicis BY anclajes;
grunt>
```

3. Contar cuántos hay en cada grupo: Ahora le decimos a Pig que cuente cuántos elementos hay en cada "saco" que creó en el paso anterior.

```
resultado = FOREACH grupo_anclajes GENERATE group, COUNT(bicis);
```

A screenshot of a terminal window titled 'cloudera@quickstart:~'. The terminal shows the execution of a Pig script. The prompt is 'grunt>'. The script being entered is: 'resultado = FOREACH grupo_anclajes GENERATE group, COUNT(bicis);', followed by a new line, and then 'grunt>' on a new line.

```
cloudera@quickstart:~
File Edit View Search Terminal Help
grunt> resultado = FOREACH grupo_anclajes GENERATE group, COUNT(bicis);
grunt>
```

4. Ver el resultado en pantalla: Para comprobar que ha funcionado antes de guardar:

DUMP resultado;

```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
  
grunt> DUMP resultado;  
2026-04-30 10:31:38,497 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP  
2026-04-30 10:31:38,498 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[Ad  
ch, ColumnMapKeyPrune, DuplicateForEachColumnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer  
TypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFi  
StreamTypeCastInserter], RULES DISABLED=[FilterLogicExpressionsSimplifier, PartitionFilterOptimizer]}  
2026-04-30 10:31:38,510 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File con  
tion threshold: 100 optimistic? false  
2026-04-30 10:31:38,513 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.CombinerOptimizer - C  
g to move algebraic foreach to combiner  
2026-04-30 10:31:38,516 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer -  
an size before optimization: 1  
2026-04-30 10:31:38,516 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer -  
an size after optimization: 1  
  
cloudera@quickstart:~  
File Edit View Search Terminal Help  
  
Job Stats (time in seconds):  
JobId Maps Reduces MaxMapTime MinMapTime AvgMapTime MedianMapTime MaxReduceTime MinReduceTime  
duceTime MedianReduceTime Alias Feature Outputs  
job_1777567184197_0006 1 1 5 5 5 7 7 7 7 bicis,grupo_anc  
resultado GROUP_BY,COMBINER hdfs://quickstart.cloudera:8020/tmp/temp1776456349/tmp-1248226650,  
  
Input(s):  
Successfully read 51 records (13790 bytes) from: "/user/cloudera/bicis/aparcamiento-bicicleta.csv"  
  
Output(s):  
Successfully stored 3 records (49 bytes) in: "hdfs://quickstart.cloudera:8020/tmp/temp1776456349/tmp-1248226650"  
  
Counters:  
Total records written : 3  
Total bytes written : 49  
Spillable Memory Manager spill count : 0  
Total bags proactively spilled: 0  
Total records proactively spilled: 0  
  
Job DAG:  
job_1777567184197_0006  
  
2026-04-30 10:32:16,798 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - S  
!  
2026-04-30 10:32:16,799 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - fs.default.name is deprecated.  
d, use fs.defaultFS  
2026-04-30 10:32:16,799 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecate  
tead, use mapreduce.jobtracker.address  
2026-04-30 10:32:16,799 [main] INFO org.apache.pig.data.SchemaTupleBackend - Key [pig.schematuple] was not set... will  
enerate code.  
2026-04-30 10:32:16,816 [main] INFO org.apache.hadoop.mapreduce.lib.input.FileInputFormat - Total input paths to proce  
2026-04-30 10:32:16,816 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.MapRedUtil - Total input paths  
cess : 1  
(tipo,1)  
("Abierto",38)  
("Cerrado",12)  
grunt>
```

4. Guardar el resultado en un archivo

STORE resultado INTO '/user/cloudera/bicis/tarea_final' USING PigStorage(';');

```

cloudera@quickstart:~
File Edit View Search Terminal Help

grunt> STORE resultado INTO '/user/cloudera/bicis/tarea_final_OK' USING PigStorage(';');
2026-04-30 10:35:55,604 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP
2026-04-30 10:35:55,684 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[Ad
ch, ColumnMapKeyPrune, DuplicateForEachColumnRewrite, GroupByConstParallelSetter, ImplicitSplitInserter, LimitOptimizer
TypeCastInserter, MergeFilter, MergeForEach, NewPartitionFilterOptimizer, PushDownForEachFlatten, PushUpFilter, SplitFi
StreamTypeCastInserter], RULES_DISABLED=[FilterLogicExpressionSimplifier, PartitionFilterOptimizer]}
2026-04-30 10:35:55,696 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File con
tion threshold: 100 optimistic? false
2026-04-30 10:35:55,697 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.CombinerOptimizer - C
g to move algebraic foreach to combiner
2026-04-30 10:35:55,698 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer -
an size before optimization: 1
2026-04-30 10:35:55,698 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer -
an size after optimization: 1
2026-04-30 10:35:55,720 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:
2026-04-30 10:35:55,722 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig script settings are added to the j
2026-04-30 10:35:55,729 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler -
.job.reduce.markreset.buffer.percent is not set, set to default 0.3

lete
2026-04-30 10:36:19,394 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 5
plete
2026-04-30 10:36:32,089 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 1
mplete
2026-04-30 10:36:32,090 [main] INFO org.apache.pig.tools.pigstats.SimplePigStats - Script Statistics:

HadoopVersion  PigVersion      UserId  StartedAt      FinishedAt      Features
2.6.0-cdh5.13.0 0.12.0-cdh5.13.0 cloudera 2026-04-30 10:35:55 2026-04-30 10:36:32 GROUP_BY

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime
duceTime  MedianReduceTime  Alias  Feature  Outputs
job_1777567184197_0007 1 1 6 6 6 6 6 6
resultado GROUP_BY,COMBINER /user/cloudera/bicis/tarea_final_OK,

Input(s):
Successfully read 51 records (13790 bytes) from: "/user/cloudera/bicis/aparcamiento-bicicleta.csv"

Output(s):
Successfully stored 3 records (33 bytes) in: "/user/cloudera/bicis/tarea_final_OK"

Counters:
Total records written : 3
Total bytes written : 33
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_1777567184197_0007

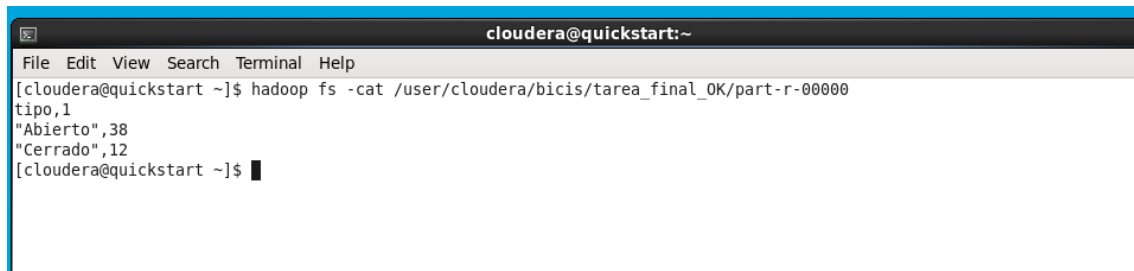
2026-04-30 10:36:32,254 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - S
!
grunt>

```

Paso 4: Cómo ver el archivo guardado

Salimos de grunt con quit y en la terminal normal de cloudera ejecutamos el comando:

hadoop fs -cat /user/cloudera/resultado_bicis/part-r-00000



```
cloudera@quickstart:~  
File Edit View Search Terminal Help  
[cloudera@quickstart ~]$ hadoop fs -cat /user/cloudera/bicis/tarea_final_0K/part-r-00000  
tipo,1  
"Abierto",38  
"Cerrado",12  
[cloudera@quickstart ~]$
```

Lo que dice el resultado:

El comando `hadoop fs -cat` nos muestra el interior del archivo que generó Pig:

- **"Abierto", 38:** Hay 38 aparcamientos de ese tipo.
- **"Cerrado", 12:** Hay 12 aparcamientos de ese tipo.
- **Total:** 50 registros (más la línea de cabecera), que coinciden con los 51 que leyó el sistema al principio.

¿Qué hemos aprendido?

Que hemos pasado por todo el ciclo de un ingeniero de datos:

1. **Ingesta:** Traer datos de fuera (`sf_datos`) al ecosistema Hadoop (`put`).
2. **Limpieza:** Ajustar el separador de carga (`;`) para que Pig entienda el archivo.
3. **Procesamiento:** Agrupar (`GROUP`) y realizar cálculos (`COUNT`).
4. **Almacenamiento:** Guardar el resultado final en un archivo persistente (`STORE`).